

Experiment No.: 01

Installation of Git and Working with Local and Remote Git Repositories

1. Objective

To install and configure **Git** and perform basic operations on **local and remote Git repositories**, such as creating a repository, adding files, committing changes, connecting to a remote repository, and pushing files to GitHub.

2. Theory

Git is a distributed version control system used to track changes in source code and other files. It helps developers maintain different versions of a project and collaborate with other developers.

A **local repository** is stored on the user's computer. It allows users to create, modify, add, and commit files locally.

A **remote repository** is stored on a remote platform such as **GitHub**. It allows users to store projects online and collaborate with other developers.

Basic Git Workflow

Working Directory → Staging Area → Local Repository → Remote Repository

- **Working Directory:** Files are created or modified here.
- **Staging Area:** Changes are selected using the `git add` command.
- **Local Repository:** Changes are permanently recorded using `git commit`.
- **Remote Repository:** Local commits are uploaded using `git push`.

3. Installation of Git

Step 1: Download Git

1. Open a web browser.
2. Visit the official Git website.
3. Download **Git for Windows**.

Step 2: Run the Installer

1. Open the downloaded Git installer.
2. Click **Next** through the installation screens.
3. Keep the default installation options.
4. Make sure **Git Bash** is selected.
5. Continue clicking **Next**.

Step 3: Complete Installation

1. Click **Install**.
2. Wait for the installation to complete.
3. Click **Finish**.

Step 4: Verify Installation

Open **Git Bash** or Command Prompt and enter:

```
git --version
```

If Git is installed successfully, the installed Git version will be displayed.

4. Configure Git

-> Set the Git username:

```
git config --global user.name "Your Name"
```

-> Set the Git email:

```
git config --global user.email "your@email.com"
```

-> Check the configuration:

```
git config --list
```

```
swwcoe@LAB01-015 MINGW64 ~/DevOps (master)
$ git --version
git version 2.55.0.windows.5

swwcoe@LAB01-015 MINGW64 ~/DevOps (master)
$ git config --global user.name "Srushti Kandakur"

swwcoe@LAB01-015 MINGW64 ~/DevOps (master)
$ git config --global user.email "shruukandakur88@gmail.com"

swwcoe@LAB01-015 MINGW64 ~/DevOps (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=true
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Srushti Kandakur
user.email=shruukandakur88@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
remote.origin.url=https://github.com/shruukandakur88-web/DevOps.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
```

5. Working with Local Git Repository

Step 1: Create a Project Folder

Create a project folder, for example:

DevOps Practicals

Open Git Bash in that folder.

Step 2: Initialize the Repository

git init

This creates a new Git repository in the project folder.

Step 3: Create a File

Create a file such as:

index.html

Add some content to the file.

Step 4: Check Repository Status

git status

This displays the current status of files in the repository

Step 5: Add File to Staging Area

To add a specific file:

git add index.html

To add all files:

git add .

Step 6: Commit the Changes

git commit -m "Initial commit"

The changes are now saved in the local repository.

```
sswcoe@LAB01-015 MINGW64 ~/DevOps (master)
$ mkdir DevOps_Practical

sswcoe@LAB01-015 MINGW64 ~/DevOps (master)
$ cd DevOps_Practical

sswcoe@LAB01-015 MINGW64 ~/DevOps/DevOps_Practical (master)
$ git init
Initialized empty Git repository in c:/Users/ask69/DevOps/DevOps_Practical/.git/

sswcoe@LAB01-015 MINGW64 ~/DevOps/DevOps_Practical (master)
$ echo This is DevOps 1st Practical, And Git is version control tool >devops.txt

sswcoe@LAB01-015 MINGW64 ~/DevOps/DevOps_Practical (master)
$ git add devops.txt
warning: in the working copy of 'devops.txt', LF will be replaced by CRLF the next time Git touches it

sswcoe@LAB01-015 MINGW64 ~/DevOps/DevOps_Practical (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   devops.txt

sswcoe@LAB01-015 MINGW64 ~/DevOps/DevOps_Practical (master)
$ git commit -m "Initial commit"
[master (root-commit) abe0d2c] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 devops.txt
```

6. Working with Remote Git Repository

Step 1: Create a GitHub Repository

1. Login to GitHub.
2. Create a new repository.
3. Enter a suitable repository name.
4. Create the repository.
5. Copy the repository URL.

Step 2: Connect Local Repository to Remote Repository

git remote add origin <repository-url of GitHub>

Example:

git remote add origin https://github.com/username/Example.git

Step 3: Check Remote Connection

git remote -v

This displays the connected remote repository.

Step 4: Rename Branch to Main

git branch -M main

Step 5: Push Files to GitHub

git push -u origin main

```
sswcoe@LAB01-015 MINGW64 ~/DevOps/Devops_Practical (master)
$ git remote add origin https://github.com/shruukandakur88-web/DevOps_Practical.git

sswcoe@LAB01-015 MINGW64 ~/DevOps/Devops_Practical (master)
$ git remote -v
origin https://github.com/shruukandakur88-web/DevOps_Practical.git (fetch)
origin https://github.com/shruukandakur88-web/DevOps_Practical.git (push)

sswcoe@LAB01-015 MINGW64 ~/DevOps/Devops_Practical (master)
$ git branch -M main

sswcoe@LAB01-015 MINGW64 ~/DevOps/Devops_Practical (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 20 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 281 bytes | 140.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shruukandakur88-web/DevOps_Practical.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

7. Commands Used

Command	Purpose
git --version	Check Git installation
git config --global user.name	Set Git username
git config --global user.email	Set Git email
git config --list	Display Git configuration
git init	Initialize a local repository
git status	Check repository status
git add	Add a specific file
git add .	Add all files
git commit -m "message"	Save changes to local repository
git branch -M main	Rename branch to main
git remote add origin	Connect local and remote repository
git remote -v	Display remote repository
git push -u origin main	Upload commits to remote repository

8. Result

Git was successfully installed and configured. A local Git repository was created, files were added and committed, and the local repository was successfully connected to a remote GitHub repository and pushed.

9. Conclusion

Thus, Git installation and configuration were successfully performed. Basic operations on **local and remote Git repositories** were carried out, including initialization, staging, committing, remote connection, and pushing files to GitHub.